10.3.2 Givens' rotations

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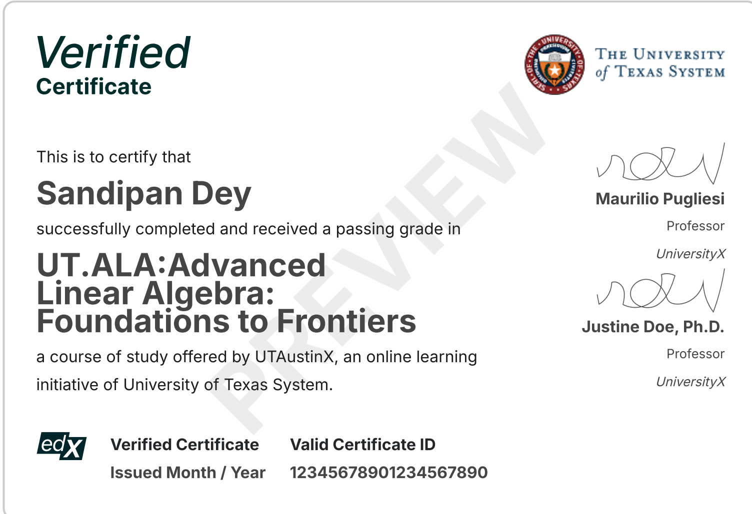

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factorization and we

didn't actually explicitly form Q -- maybe we can take those transformations and

apply them one by one as opposed to forming Q. And notice that it's obviously

important that we use only unitary matrices to do this. Now, what kind of

unitary matrices are at our disposal? Well, we could just say, "Ah, we can compute

a Householder transformation just off of these two entries thereby introducing a

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2.0x

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Reading assignment

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Homework 10.3.2.1

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Propose formulas for γ and σ such that

$$\begin{pmatrix} \gamma & -\sigma \\ \sigma & \gamma \end{pmatrix}^T \underbrace{\begin{pmatrix} \chi_1 \\ \chi_2 \end{pmatrix}}_{x} = \begin{pmatrix} \|x\|_2 \\ 0 \end{pmatrix},$$

where $\gamma^2 + \sigma^2 = 1$.

- ( ) $\gamma = \frac{\chi_1}{\|x\|_2}$ and $\sigma = -\frac{\chi_2}{\|x\|_2}$
- ( ) $\gamma = -\frac{\chi_1}{\|x\|_2}$ and $\sigma = \frac{\chi_2}{\|x\|_2}$
- (•) $\gamma = \frac{\chi_1}{\|x\|_2}$ and $\sigma = \frac{\chi_2}{\|x\|_2}$
- ( ) $\gamma = -\frac{\chi_1}{\|x\|_2}$ and $\sigma = -\frac{\chi_2}{\|x\|_2}$

✓

Solution

For a complete solution, see the notes .

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